

Database Normalization Process

Step 1:

- Present data in tabular format, where each cell has a single value
- Eliminate repeating group & nulls by making sure that each attribute contains an appropriate data value
- Identify the Primary Keys

Step 2:

- Identify All Dependencies
- Remove partial dependency

Step 3:

- Remove Transitive Dependencies

Un-Normalize Form



1st Normal Form



2nd Normal Form



3rd Normal Form

Given the un-normalized form (UNF), normalized it into third normal form (3NF)

Proj. Num	Project Name	Employee Number	Employee Name	Job Class	Chg/ Hour	Hours Billed	Total Charge
1	Hurricane	101	John	Elect.Eng	65	13	845
		102	David	Comm.Tech	60	16	960
		104	Anne	Comm.Tech	60	19	1140
		Subtotals				2945	
2	Coast	101	John	Elect.Eng	65	15	975
		103	June	Biol.Eng	55	17	935
		Subtotals				1910	
3	Satellite	104	Anne	Comm.Tech	60	18	1080
		102	David	Comm.Tech	60	14	840
		Subtotals				1920	

Present the data in table form, fill up all rows and columns. Ensure each row and intersection has only one value

ProNum	ProName	EmpNum	EmpName	JobClass	ChgHour	HourBill	TotChar	SubTot
1	Hurricane	101	John	Elect.Eng	65	13	845	2945
1	Hurricane	102	David	Comm.Tech	60	16	960	2945
1	Hurricane	104	Anne	Comm.Tech	60	19	1140	2945
2	Coast	101	John	Elect.Eng	65	15	975	1910
2	Coast	103	June	Biol.Eng	55	17	935	1910
3	Satelite	104	Anne	Comm.Tech	60	18	1080	1920
3	Satelite	102	David	Comm.Tech	60	14	840	1920

Identify Primary keys: ProNum&EmpNum

The table is now in first normal form (1NF)

Change to second normal form (2NF)

Primary keys: **ProNum&EmpNum** (in most cases, there will be more than 1 attributes known as composite PK)

Identify all dependencies. Remove partial dependency (exists when table has composite PK)

Make sure that each non-key attributes / non-determinant / non-PK is fully dependent on its determinant (PK)

Composite PK	Non-key attributes	Check List	Action	
ProNum&EmpNum	ProName	Is ProName fully dependent on both (ProNum&EmpNum)? Answer: No , ProName depend on ProNum, not EmpNum	Partial dependency exists. Remove ProName together with its determinant (ProNum) to form a new table.	Form new table
ProNum&EmpNum	SubTot	Is SubTot fully dependent on both (ProNum&EmpNum)? Answer: No , SubTot depend on ProNum, not EmpNum	Partial dependency exists. Remove SubTot together with its determinant (ProNum) to form a new table.	
ProNum&EmpNum	EmpName	Is EmpName fully dependent on both (ProNum&EmpNum)? Answer: No , EmpName depend on EmpNum, not ProNum	Partial dependency exists. Remove EmpName together with its determinant (EmpNum) to form a new table.	Form new table
ProNum&EmpNum	JobClass	Is JobClass fully dependent on both (ProNum&EmpNum)? Answer: No , JobClass depend on EmpNum, not ProNum	Partial dependency exists. Remove JobClass together with its determinant (EmpNum) to form a new table.	
ProNum&EmpNum	ChgHour	Is ChgHour fully dependent on both (ProNum&EmpNum)? Answer: No , ChgHour depend on EmpNum, not ProNum	Partial dependency exists. Remove ChgHour together with its determinant (EmpNum) to form a new table.	
ProNum&EmpNum	HourBill	Is HourBill fully dependent on both (ProNum&EmpNum)? Answer: Yes , HourBill depend on both (ProNum&EmpNum)	Full dependency exists, therefore no action needed. HourBill remains in the original table with its determinant (ProNum&EmpNum)	Remain at original table
ProNum&EmpNum	TotChar	Is TotChar fully dependent on both (ProNum&EmpNum)? Answer: Yes , TotChar depend on both (ProNum&EmpNum)	Full dependency exists, therefore no action needed. TotChar remains in the original table with its determinant (ProNum&EmpNum)	

Tables in second normal form (2NF)

ProNum	ProName	SubTot
1	Hurricane	2945
2	Coast	1910
3	Satelite	1920

ProNum	EmpNum	HourBill	TotChar
1	101	13	845
1	102	16	960
1	104	19	1140
2	101	15	975
2	103	17	935
3	104	18	1080
3	102	14	840

EmpNum	EmpName	JobClass	ChgHour
101	John	Elect.Eng	65
102	David	Comm.Tech	60
103	June	Biol.Eng	55
104	Anne	Comm.Tech	60

The tables are now in second normal form (2NF)

Convert to third normal form (3NF)

Identify transitive dependency (if any) then remove it, else the tables are already in third normal form (3NF)

Transitive dependency occurs when a non-key attribute is dependent on another non-key attribute

In this example, transitive dependency exists: EmpNum > JobClass > ChgHour (ChgHour is transitively dependent on EmpNum)

ChgHour (which is a non-key attribute) is dependent on JobClass (which is also a non-key attribute), because JobClass determines ChgHour

Therefore, remove JobClass & ChgHour to form a new table. However, JobClass need to remain in Employee table as a reference

Tables in third normal form (3NF)

ProNum	ProName	SubTot
1	Hurricane	2945
2	Coast	1910
3	Satelite	1920

EmpNum	EmpName	JobClass
101	John	Elect.Eng
102	David	Comm.Tech
103	June	Biol.Eng
104	Anne	Comm.Tech

ProNum	EmpNum	HourBill	TotChar
1	101	13	845
1	102	16	960
1	104	19	1140
2	101	15	975
2	103	17	935
3	104	18	1080
3	102	14	840

JobClass	ChgHour
Elect.Eng	65
Comm.Tech	60
Biol.Eng	55

The tables are now in third normal form (3NF), normalization process ends.

Now you can implement the database in the DBMS, you may make further changes to the tables in 3NF such as below

Implement the tables in database server

You can make further changes by adding/modifying PK & FK

Project

ProNum (PK)	ProName	SubTot
1	Hurricane	2945
2	Coast	1910
3	Satelite	1920

ProjectWork

ID (PK)	ProNum (FK)	EmpNum (FK)	HourBill	TotChar
01	1	101	13	845
02	1	102	16	960
03	1	104	19	1140
04	2	101	15	975
05	2	103	17	935
06	3	104	18	1080
07	3	102	14	840

Employee

EmpNum (PK)	EmpName	JobClassID (FK)
101	John	J01
102	David	J02
103	June	J02
104	Anne	J03

Rate

JobClassID (PK)	JobClass	ChgHour
J01	Elect.Eng	65
J02	Comm.Tech	60
J03	Biol.Eng	55